

Department Overview

- **Department Name:** Department of Electronic Engineering
- **Faculty:** Faculty of Information & Communication Technology (FICT)
- **Accreditation:** The program is fully accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Technological Focus:** Robotics, Artificial Intelligence, Digital Manufacturing, Automated and Electric Vehicles, Information Security, Wireless Networks, and Nano-Technologies.
- **History & Impact:** Established in 2003, the department focuses on meeting the shifting technological needs of industry and society, playing a primary role in technical education within Balochistan.
- **Career Fields:** Graduates are employed across diverse sectors, including national and international research institutions, telecom companies, banking, automotive, textile, biomedical, petrochemical, manufacturing, freelancing, and entrepreneurship.

Program Educational Objectives (PEOs)

The department aims to produce electronic engineers who can:

1. **PEO-1:** Contribute their knowledge for the betterment of humanity in general and society in particular.
2. **PEO-2:** Impart their analytical and innovation skills for the enhancement of research, development, and production of Electronic Systems.
3. **PEO-3:** Work efficiently in teams or individually, integrating effectively in diverse cultures and environments.

Bachelor of Science (BS) Program

Admission Requirements

- **F.Sc Pre-Engineering:** Completed from any recognized board or equivalent with a minimum of **60% marks**.
- **Diploma of Associate Engineer (DAE):** Completed in a relevant field with at least **60% marks**.

Degree Requirements

- **Total Credit Hours:** 136
- **Total Courses:** 45
- **Minimum CGPA:** ≥ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour listings follow the format (Theory + Lab).

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	HUM-163 /	Functional	NIL / NIL	3+0 / 3+0

	MATHP-118	English / Calculus and Analytical Geometry		
	EE-112 & EE- 112L	Linear Circuit Analysis	NIL	3+1
	PHY-205 & PHY-205L	Applied Physics	NIL	2+1
	CS-123 / ELE- 102L	Applications of ICT / Workshop Practice	NIL / NIL	2+1 / 0+1
2nd	HUM-268 / HUM-101	Communicatio n Skills / Islamic Studies	NIL / NIL	3+0 / 2+0
	EE-208 & EE- 208L	Electronic Devices & Circuits	NIL	3+1
	ENGG-217L / HUM-102	Engineering Drawing / Pakistan Studies	NIL / NIL	0+1 / 2+0
	MATHP-111	Linear Algebra	NIL	3+0
3rd	STAT-403 / MATHA-214	Probability & Random Variables / Differential Equations	NIL / Calculus, Geometry, & Linear Alg	3+0 / 3+0

	EE-110 & EE-110L	Digital Logic Design	NIL	3+1
	EE-204 & EE-204L	Instrumentation & Measurements	Electronic Devices & Circuits	2+1
	EE-109 & EE-109L	Electronic Circuit Design	NIL	3+1
4th	HUM-276 / MATHA-213	TRW & PS / Complex Variables and Transform	NIL / NIL	3+0 / 3+0
	EE-491 & EE-491L	Analog Integrated Electronics	Electronic Circuit Design	2+1
	EE-201 & EE-201L	Electrical Network Analysis	Linear Circuit Analysis	3+1
	CE-324 & CE-324L	Microprocessors and Microcontrollers	Digital Logic Design	3+1
5th	EE-324 & EE-324L	Electrical Machines	NIL	3+1
	STAT-403	Signal & Systems	NIL	2+1
	EE-241 & EE-241L	Control Systems	Linear Algebra	3+1

	CS-212 & CS-212L	Object Oriented Programming for Engineers	NIL	3+1
	TE-301 & TE-301L	Data Communication & Networks	Communication Systems	2+1
6th	EE-421	Power Electronics	Electrical Machines	3+1
	MATHA-216 & L / EE-XX	Numerical Analysis & Lab / IDEE-I	— / —	2+1 / 3+0
	TE-314 & TE-314L	Communication Systems	Signals and Systems	3+1
	MGMT-XX	Management Elective-I	NIL	3+0
7th	EE-XX / EE-XX	Elective-I / Elective-II	NIL / NIL	3+0 / 3+0
	MGMT-XX	Management Elective-II	NIL	3+0
	EE-441 & EE441L	Industrial Electronics	NIL	3+1
	XX-XX / EE-481L	Social Science Elective-I / Final Year Project Part-I	NIL / NIL	2+0 / 0+3

8th	EE-XX / EE-XX	Elective-III / Elective-IV	NIL / NIL	3+0 / 3+0
	TE-412 & TE-412L	Digital Signal Processing	Signals & Systems	2+1
	XX-XX / XX-XX	Social Science Elective-II / IDEE-II	NIL / NIL	2+0 / 3+0
	EE-482L	Final Year Project Part-II	Final Year Project Part-I	0+3

Master of Science (MS) Program

Admission Requirements

- **Academic Background:** BE/BS degree in Electronic, Electrical, Computer, or Telecommunication Engineering from an HEC-recognized institution.
- **Academic Performance:** Minimum of **60% marks** under the annual grading system or a minimum CGPA of **2.5 out of 4.0**.
- **Standardized Testing:** A valid **GAT General** score with at least a **50%** cumulative result.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Courses:** 8
- **Minimum Academic Standing:** CGPA $\geq$ 2.5